

TOAMASINA, Madagascar

WMO: 670950

Lat: 18.110S

Lon: 49.393E

Elev: 7

StdP: 101.24

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	17.2	15.0	10.6	19.5	15.8	11.2	19.6	9.4	22.9	8.4	22.8	0.9	240	0.280

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	31.8	26.5	31.1	26.5	30.8	26.3	27.5	30.3	27.1	30.0	26.7	29.6	4.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.5	29.8	26.2	21.6	29.2	26.0	21.4	29.0	87.7	30.3	85.8	30.5	84.2	29.7	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.3	DB	15.1	34.9	0.6	2.0	14.7	36.3	14.3	37.5	14.0	38.6	13.5	40.0	(4)
(5)			WB	14.4	28.7	0.7	0.6	13.9	29.1	13.5	29.5	13.1	29.9	12.5	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	27.1	27.1	26.8	25.8	24.3	22.6	21.5	21.7	22.3	23.7	25.1	26.5	(6)	
	DBStd	2.31	0.92	1.02	0.91	0.99	1.18	1.17	1.02	0.96	0.97	1.20	1.15	0.92	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	5307	530	480	522	475	442	378	357	364	370	424	452	513	(10)	
	CDD18.3	2265	272	247	264	225	183	128	99	106	120	166	202	254	(11)	
	CDH23.3	17380	2744	2485	2479	1752	1059	482	270	354	533	1069	1657	2496	(12)	
	CDH26.7	4551	919	854	777	476	194	34	4	4	15	120	359	792	(13)	
(14) Wind	WSAvg	2.4	2.3	2.4	2.3	2.3	2.3	2.6	2.9	2.5	2.2	2.3	2.1	2.1	(14)	
(15) Precipitation	PrecAvg	3236	416	363	461	295	266	260	284	201	123	115	159	300	(15)	
	PrecMax	4456	1015	867	1194	978	1476	534	551	431	287	395	419	909	(16)	
	PrecMin	2327	78	53	109	65	69	82	97	75	34	20	16	18	(17)	
	PrecStd	569	207	206	224	165	207	103	109	81	66	85	100	201	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	32.9	32.1	31.1	30.0	28.2	27.1	27.1	27.8	29.2	30.3	31.9	(19)	
		MCWB	26.4	26.7	26.4	26.1	25.3	23.8	22.4	22.3	22.8	24.4	25.7	26.3	(20)	
	2%	DB	31.4	31.8	31.1	30.2	29.1	27.7	26.2	26.3	27.1	28.2	29.8	31.1	(21)	
		MCWB	26.6	26.7	26.4	25.7	24.7	23.3	21.7	21.7	22.3	23.9	25.4	26.4	(22)	
	5%	DB	31.0	31.1	30.8	29.9	28.5	27.0	25.8	26.0	26.6	27.9	29.1	30.3	(23)	
		MCWB	26.5	26.5	26.3	25.5	24.3	22.7	21.4	21.5	22.1	23.7	24.8	25.9	(24)	
	10%	DB	30.2	30.7	30.1	29.1	27.9	26.1	25.0	25.2	26.0	27.2	28.6	30.0	(25)	
		MCWB	26.1	26.4	25.9	25.1	24.0	22.1	21.1	21.0	21.8	23.0	24.4	25.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.1	27.8	27.1	26.2	25.0	23.6	23.4	23.8	25.7	26.7	27.6	(27)	
		MCDB	30.7	30.8	30.2	29.7	28.6	27.5	25.8	26.0	26.5	28.2	29.4	30.3	(28)	
	2%	WB	27.4	27.5	27.2	26.5	25.6	24.0	22.7	22.6	23.1	24.9	26.1	27.0	(29)	
		MCDB	30.2	30.4	29.9	29.3	28.2	26.6	25.2	25.4	26.0	27.4	28.7	30.0	(30)	
	5%	WB	27.0	27.0	26.7	26.0	24.9	23.4	22.1	22.1	22.6	24.3	25.5	26.6	(31)	
		MCDB	29.9	30.1	29.6	29.0	27.6	26.0	24.5	24.8	25.6	26.9	28.2	29.5	(32)	
	10%	WB	26.6	26.6	26.4	25.6	24.4	22.7	21.5	21.6	22.2	23.7	25.0	26.1	(33)	
		MCDB	29.4	29.6	29.2	28.4	27.1	25.3	24.0	24.3	25.1	26.4	27.7	29.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.8	6.7	6.8	7.0	6.9	7.1	7.4	7.6	7.4	7.2	6.9	(35)	
		MCDBR	7.6	7.6	7.5	7.4	7.9	8.0	8.3	8.2	8.3	7.6	7.5	7.4	(36)	
	5% WB	MCWBR	3.7	3.8	3.7	3.8	4.2	4.2	4.5	4.4	4.4	4.0	3.9	3.8	(37)	
		MCWBR	6.9	7.1	6.8	6.8	7.2	7.2	7.3	7.5	7.6	6.9	6.8	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.398	0.403	0.392	0.372	0.362	0.349	0.348	0.351	0.371	0.396	0.406	0.400	(40)	
		taud	2.549	2.536	2.568	2.614	2.625	2.625	2.614	2.580	2.503	2.446	2.445	2.509	(41)	
	Ebn at Noon		947	932	920	903	876	871	882	910	926	928	933	944	(42)	
		Edh at Noon	110	110	103	92	85	82	85	94	108	119	121	115	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.87	5.69	5.22	4.71	3.90	3.54	3.60	4.37	5.30	5.88	6.35	6.47	(44)	
	RadStd		0.47	0.46	0.35	0.33	0.27	0.18	0.24	0.17	0.25	0.32	0.38	0.44	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.39	N/A	N/A	+0.33	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page